

Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

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Abstract

Rac1 (Ras-related C3 botulinum toxin substrate 1) serves as a molecular switch that hydrolyses GTP to GDP. Due to the slow hydrolysis and dissociation of GDP, Rac1 requires Guanine Exchange Factors (GEFs) and GTPase-activating proteins (GAPs) for regulation. Despite being considered undruggable because of its small size and shallow surface, the binding of GAPs and GEFs suggests the existence of transient binding pockets on Rac1. We identified four such pockets: two serve as GEF binding sites, a third interacts with DHR domain-containing GEFs from the DOCK family, and the fourth binds the Engulfment and Cell Motility 1 (ELMO1) protein. To adequately sample the open states of these pockets, we used cosolvent probes in MD simulations. Principal component analysis of the cosolvent MD simulations helped us understand the domain movements that cause these transient pockets to open and close.

Introduction

Small G proteins are a ubiquitous class of proteins that play pivotal roles in various cellular processes. Their functions include signal transduction, vesicle trafficking, cytoskeletal dynamics, and cell division. Ras GTPases regulate cell proliferation, differentiation, and survival. Rho GTPases govern cytoskeletal dynamics and cell migration. These proteins possess conserved domains across several families; they are made up of six β -sheets surrounded by 5 α -helices. All domains are conserved such that they carry a guanine nucleoside binding site to recognise the guanine base and phosphate groups and the magnesium ion.^{1,2} The switch regions regulate the traffic to the guanine nucleoside binding site. Their activity is regulated through a molecular switch mechanism by guanine nucleotide exchange factors (GEFs), GTPase-activating proteins (GAPs), and guanine nucleotide dissociation inhibitors (GDIs).² The mechanism cycles between OFF (GDP-bound inactive) and ON (GTP-bound active) states. This switching mechanism controls various processes like cell growth and differentiation to vesicular and nuclear transport.^{2,3} It is also well established that small G proteins alone cannot hydrolyse GTP effectively. Therefore, GEFs and GAPs are known to assist in the catalytic function and thus the switching mechanism. They control the dynamics of the switch regions that facilitate either the hydrolysis of GTP or the dissociation of GDP. GDP:Mg²⁺ dissociates from GTPases very slowly, and therefore, GEFs assist in exchanging GDP for GTP. One of the accepted mechanisms of GEFs in the dissociation of GDP is by electrostatic repulsion of the magnesium ion that forms a strong interaction with small G proteins and phosphate groups of GDP. The loss of a magnesium ion weakens the affinity of GDP, which eventually dissociates. Alternatively, some GEFs can rearrange the switch I or II conformations to reduce GDP's affinity, thus facilitating its dissociation. However, GEFs can also combine the two mechanisms to enable GDP dissociation.^{4,5} Furthermore, GAPs stabilise the transition state for GTP:Mg²⁺:small G proteins, and facilitate the hydrolysis of the γ -phosphate group from GTP. Taken together, the switching mechanism can be seen as GEFs assist in the activation of small G proteins, while GAPs assist in GTP hydrolysis. Following GTP hydrolysis, the inactive small G proteins prevent further signal transduc-

tion.⁶ In some classes of small G proteins, the active and inactive states are conjugated with small G proteins being shuttled between the cell membrane and cytosol. Such small G proteins undergo post-translational modifications to carry long-chain fatty acids at their C-terminal to assist bilayer insertion. GDIs^{7,8} sequester the lipid component to prevent membrane insertion of small G proteins in the cytosol. This coordinated action of small G proteins, GEFs, GAPs, and GDIs is essential for various cellular processes. Dysregulation of these components contributes to pathologies like cancer, neurodegeneration, and inflammatory disorders. For instance, mutations in Ras and its regulators are implicated in oncogenesis, highlighting the therapeutic potential of targeting small G protein signalling pathways.² Such mutations (Q61L, Y72C, oncogenic P29S and Q61R, and negative T17N) are known to disrupt the equilibrium between the active and inactive states of small G protein.⁹ It has been observed that mutations either enhance the affinity of GEFs or GAPs that trap the small G proteins either in an activated state or an inactivated state. Small molecule inhibitors targeting small G protein activity or disruptors of GEF/GAP interactions¹⁰ are under development for cancer therapy. However, there are several challenges in targeting small G protein and/or their GAP/GEF complexes. Firstly, targeting the guanine binding site is challenging owing to the large intracellular concentration of GTP and high affinity.¹¹ In such cases, finding alternate sites can avoid such competition with endogenous substrates. GTPase hyper-activation or inactivation can be corrected by disrupting the protein-protein interaction (PPI) between small G protein and GEFs/GAPs by targeting the regulatory protein docking sites.

Rac1 plays a crucial role in regulating cellular processes owing to its switching mechanism that recruits several proteins upstream and downstream. PREX1, TIAM1, VAV1 and DOCK2 are Rac1-specific GEFs that regulate many cellular processes and, therefore, are implicated in many Rac-dependent pathologies such as asthma and cancer. Structurally, PREX1, TIAM1 and VAV1 are DH/PH domain-containing proteins, while DOCK2 contains a DHR domain. Irrespective of their structural diversity, GEFs interact with the switch I and/or II region residues. Since the nucleotide-binding site is well conserved among the small G protein family, this poses a significant challenge to selec-

tively target Rac1 by modulating the nucleotide-binding site. Furthermore, the molar concentration with picomolar binding affinity of GTP makes it more difficult for small molecules to competitively inhibit Rac1. The binding of GEFs with Rac1 shows the possibility of alternative binding sites that transiently open to allow this protein-protein interaction. Therefore, the switching mechanism offers a conformational landscape that must be studied in depth to locate other binding sites on the Rac1 surface. Therefore, we embarked to understand the complete dynamical landscape of Rac1 protein bound to either nucleotides (GDP or GTP) or GEFs (PREX1, TIAM1, VAV1 or DOCK2). In this study, we determined the evolution of Rac1 binding pockets in its nucleotide-bound conformation and bound with its partners (holo). Following molecular dynamics simulations, we identified and measured the pocket volume changes for the transient pockets. We also monitored the effect of binding of various GEFs and nucleotide on the changes in the pocket volume. Lastly, using Pearson’s correlation coefficient, we evaluated the possibility of correlation between changes in all four transient pockets among each other.

Methods

Preparation of systems for molecular dynamics simulation

The coordinates for all complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in **Table 1**. The X-ray structures were imported and processed in the UCSF Chimera v1.17.¹² Any missing loops were built using Modeller¹³ with UCSF Chimera v1.17 as the GUI.^{14,15}

Table 1: Rac1 conformations and binding partners used in this study

PDB id	Rac1 Modulator/Ligand	Comments
1HE1 ¹⁶	GDP:Mg ²⁺	Inactive Rac1
3TH5 ¹⁷	GTP:Mg ²⁺	Active Rac1
4YON ¹⁸	PREX1 (aa. 34-366)	DBL family member
1FOE ¹⁹	TIAM1 (aa. 1033-1406, mouse)	DBL family member
2VRW ²⁰	VAV1 (aa. 170-575, mouse)	DBL family member
2YIN ²¹	DOCK2 (aa. 1192-1622)	DOCK family member

Generating AMBER-compatible parameters

All systems were prepared for MD simulations with the Leap module in AMBERtools²³. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field.²² Amber-compatible parameters for all nucleotides were obtained from the Manchester AMBER database.²³ For the VAV1 zinc finger region, the parameters of the Zinc atom, coordinating cysteines and histidine residues, the force field parameters were obtained from the ZAFF²⁴ force field library.

Molecular dynamics simulations

All complexes were solvated with TIP3P²⁵ waters enclosed in a box with dimensions extending 10 Å beyond any protein atom, and an appropriate number of neutralising ions were added to maintain the system’s neutrality. The complexes were relaxed using three cycles of minimisation of 50000 steps each. The first 5000 steps of minimisation adopted the steepest descent algorithm, and the remaining 45000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0 Å for the minimisation phase. The first cycle of minimization was performed with 25 kcal/(molÅ²) restraining force on the solute atoms, this force constant was reduced to 5 kcal/(molÅ²) in the next cycle, and the final cycle was performed without any restraining forces on either the solute or the solvent atoms. The system was heated in 4 stages (see **Table 2**) from 0 K to 300 K for 10 ns with 50 kcal/(molÅ²) restraining force on the solute atoms using the Langevin dynamics, with a collisional frequency of 2 ps⁻¹. The next step employed the NPT ensemble for density equilibration for another 2 cycles of 10 ns each, with the restraining force on the solute atoms gradually decreasing from 5 to 0 kcal/(molÅ²). In the equilibration phase, the cut-off for evaluating non-bonded interactions was placed at 8 Å. In the production phase of the MD simulations, all systems were simulated for 100 ns in triplicate, amounting to a 300 ns trajectory for each system. The long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) summation method.²⁶ The cut-off for evaluating non-bonded interactions was placed at 9.0 Å for the production phase. All the computations were done using the CUDA-enabled

PMEMD code^{27–29} in AMBER22.^{30,31} All simulations were performed with the SHAKE algorithm³² to constrain bonds to hydrogen atoms, with the tolerance set to 0.00001 Å. A 2 fs integration time step was used to solve Newton’s equations of motion. For analysis, the conformations were saved every 100 ps, amounting to 1000 snapshots per trajectory (3000 per system).

Table 2: Details of stages of Heating using NVT Ensemble

Stage	Temperature range (K)	Time (ns)
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
Stage 4	300 to 300	7.5 to 10

Measuring the structural dynamics of Rac1

Analysis of the MD trajectory was performed with the cpptraj³³ module in AMBER-Tools23. Before the analysis, all solvent atoms and neutralising ions were stripped off, and the protein conformations were aligned to their respective X-ray conformation. The mass-weighted C α root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The root-mean-square fluctuations (RMSF) per residue for the backbone and side-chain atoms were computed to understand the domain-specific fluctuations.

Identification and measurement of alternate pockets on Rac1

POVME2^{34,35} was employed for identifying various pockets on the Rac1. To be able to incorporate various experimental conformations of Rac1, we employed Rac1 with different binding partners (see **Table 1**). To identify the pockets, the pocketID module on POVME2 was used on the aforementioned Rac1 structures. The parameters used for pocket identification are shown in **Table 3**. Following the pocket identification, we monitored the pocket volume changes of these pockets throughout MD trajectories. It has been observed that the pocket might be a good candidate for druggability if its volume range was found to lie between 610Å³ to 930Å³.^{36–38}

Table 3: Parameters used for pocket identification

Parameter	Value
pocket detection resolution	4.0 Å
pocket measuring resolution	1.0 Å
clashing cutoff	3.5 Å
number of neighbors	4
number of spheres	5
sphere padding	5.0 Å

MD simulations with Cosolvent probes

MD simulations with cosolvent probes are very useful to accelerate pocket opening or to keep the transient pockets open for a longer timescale in MD simulations to better understand pocket characteristics. We identified transient pockets in response to GEF binding, which were otherwise not observed or very hard to locate. To ensure that these transient pockets can remain open sufficiently long to be druggable, we used cosolvent MD simulations. For this purpose, we used 1M concentration of isopropanol, phenol and toluene. The choice of these solvents mimics drug-like fragments with decreasing water solubility. In this case isopropanol ($\log P = 0.05$) > phenol ($\log P = 1.39$) > toluene ($\log P = 2.56$). The simulation box with Rac1 protein (PDB: 3TH5, GTP and Magnesium ions were deleted) and 0.15M of NaCl was prepared using packmol.³⁹ The protocol for MD simulations is the same as mentioned above. GAFF2 parameters were assigned to the organic solvents. For cosolvent MD simulations, we performed triplicate MD simulations for 1 μ s, amounting to a total sampling time of 3 μ s per solvent system. For the baseline calculations, we also prepared two systems without adding cosolvents. One is the RAC1 complexed with GTP and Magnesium ions, which mimics an active-like conformation of RAC1. The apo system was created by deleting GTP and Magnesium ions from the active site system described previously. These systems would enable us to understand if any of the transient pockets can spontaneously open without the influence of the cosolvents and any ligand in the nucleotide pocket.

Principal component analysis of Rac1 protein following cosolvent MD

To understand the effect of cosolvent probes on the Rac1, and thus opening of the transient pockets, we extracted the essential dynamics using Principal Component Analysis (PCA). The Variance-Covariance matrix for C α atoms was extracted in the bio3D package⁴⁰ module in R Studio. We extracted and analysed four principal components; however, the first principal component with the largest eigenvalue was used to understand the essential dynamics in the Rac1 protein. The scree plot for each system is given in the supplementary figure S3

Quantifying the GEF binding hotspots on Rac1 surface

The free energy of binding was computed using the equation (Eq-1) on the conformations saved from the MD simulations using the MMPBSA module⁴¹ available in AMBER-Tools23;

$$\Delta G_{\text{binding}} = \Delta E_{\text{MM}} + \Delta H_{\text{GBSA}} - T\Delta S \quad (1)$$

Where ΔE_{MM} is the interaction energy between the binding partners computed in the gas phase using a molecular mechanics method. The solvation-free energy or the enthalpy of hydration ΔH_{GBSA} was computed using the Generalised-Born Surface Area (GBSA) method, and the entropic component of binding $T\Delta S$ was not computed due to the huge computation cost. The internal dielectric constant for GBSA calculations was set to 1, and the solvent dielectric constant was fixed at 80. For computing the free energy of binding using the MMGBSA method, the OBC’s GB model⁴² ($igb = 5$) was used. The mbondi2 radii set was employed to define the PB radii for all solute atoms during the preparation of the topology and parameter files in the leap module in AMBERTools23. The ionic strength for GBSA calculations was maintained at 150 mM. All energy components, in an implicit solvent, were calculated using a large cut-off of 999 Å. The surface area for the non-polar solvation term was computed using

the linear combination of pairwise overlaps (LCPO) method.⁴³ To identify the critical amino acid residues that are critical for the protein-protein interactions between Rac1 and GEFs, we computed the per-residue decomposition of the binding affinity value. This will also enable us to estimate the contribution from the residues lining the newly identified transient pocket and their energetic contribution to GEF binding.

Results and discussion

Protein-protein interactions are central to biology. By coordinating an intricate network of proteins that are recruited and activated in a pathway to pass the signal to the nucleus to control the rate and amount of protein expression. Such networks are tightly regulated by molecular switches, mainly by small G proteins, which are activated in response to GTP binding and turned off upon GTP hydrolysis to GDP. For the next switch on step, GDP must dissociate, which allows for GTP binding. However, the binding affinity of GDP and GTP are very high and comparable; therefore, GDP dissociation is not spontaneous. Thus, the guanine exchange step must be assisted by GEFs that are recruited depending on the pathway. The exchange factors serve a dual role of nucleotide exchange and recruiting proteins downstream in the signal pathway by activating a specific GEF-dependent network of proteins. Therefore, we attempt to understand the Rac1 dynamics bound to different GEFs, nucleotides, and *apo* states. Further, the Rac1-GEF binding paves the way to unlock different binding sites on the Rac1 surface that are not normally open in the *apo* state. We further investigate whether such sites are amenable to small-molecule targeting by measuring the pocket volume changes over the course of MD trajectories. We also try to explain the structural changes that Rac1 undergoes during the opening of these alternate sites, and if there is any noticeable change in the nucleotide binding site that may affect the GTP association or GDP dissociation.

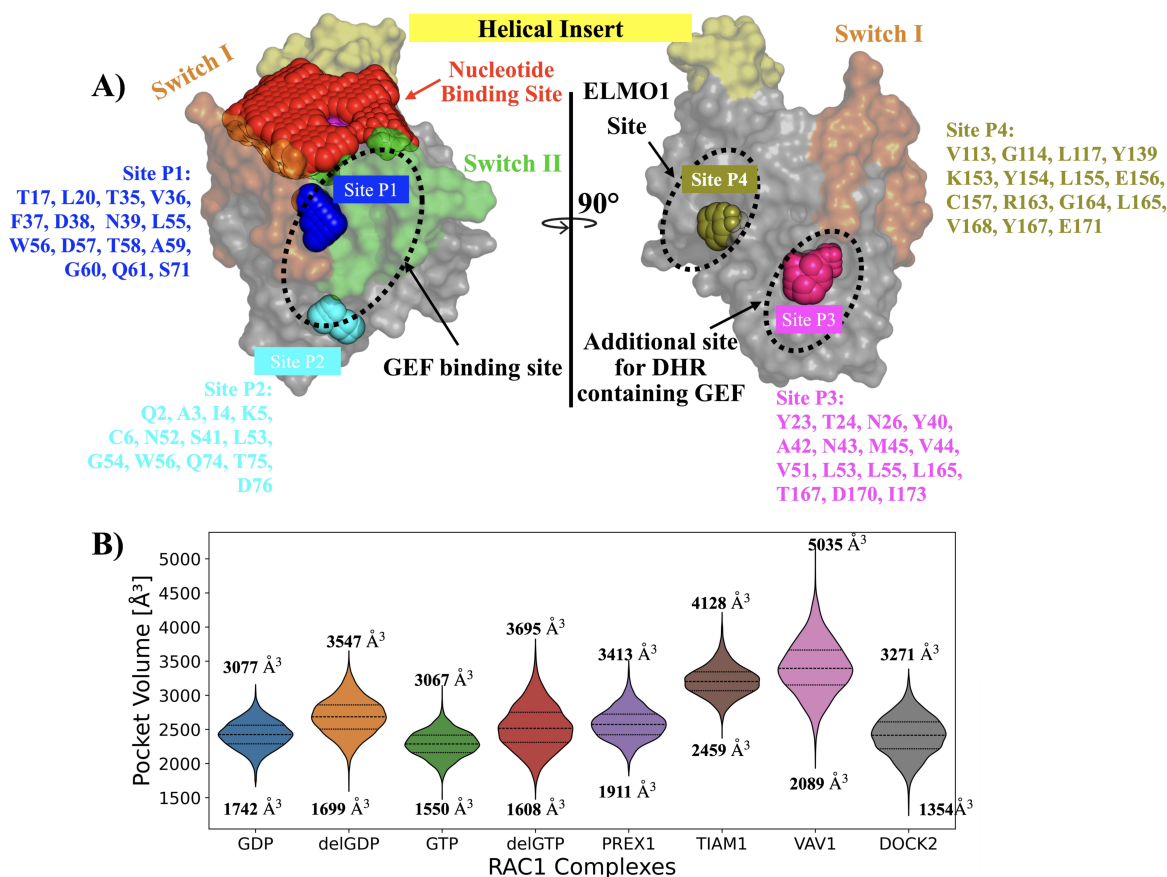


Figure 1: **A)** Rac1 surface with all possible binding pockets. The yellow-coloured region depicts the helical insert, which is peculiar to the Rho GTPase family. The orange-colored region depicts switch I, and the green colored region is switch II. Red spheres occupy the nucleotide-binding site. The pink-colored region is the P-loop. Blue spheres depict Site P1, cyan spheres show Site P2. Magenta spheres occupy the Site P3, and asparagus spheres show the Site P4. **B)** The violin plot illustrates the distribution of the pocket volumes for each of the nucleotide-binding sites. The changes in the pocket volumes are monitored when Rac1 is complexed with either GEFs, GDP, GTP or in the *apo*-states named as delGDP (GDP deleted from GDP-bound system), and delGTP (GDP deleted from GTP-bound system). The amino acid numbering for Rac1 follows the Uniprot ID: P63000. The numerical values on the violin plots indicate the minimum and maximum pocket volumes measured in the MD trajectories.

GEF docking sites open transient pockets on the Rac1 surface

We employed the POVME2³⁵ tool to identify and measure the pocket volumes of new transient sites. By definition, transient or cryptic pockets are observed only in holo proteins, while being undetectable in apo proteins. Therefore, we used various Rac1 structures bound to GEFs and nucleotide (**Table 1**). To measure the pocket volume changes of these pockets, we simulated all Rac1 complexes(**Table 1**). Before measuring the pocket volumes, all binding partners, including nucleotides, were deleted from the trajectory. In other words, the Rac1 conformation was allowed to evolve in the presence of

the binding partners, which allows opening of the transient sites. Firstly, we observed that the change in the nucleotide pocket volume was influenced by GEFs and the nucleotides. We observed that the pocket volume fluctuations remained under $3000\text{\AA}^3$ in the presence of GTP, and GDP (**Figure 1B**). This indicates the closing of the nucleotide pocket in preparation for GTP hydrolysis (GTP-bound). For the GDP-bound structure, this would indicate a state after the GTP hydrolysis. The maximum and minimum pocket volumes for nucleotide sites are similar for both GTP- and GDP-bound Rac1. Next, we monitored the changes in the pocket volumes by deleting the GTP and GDP to create *apo*-like states called "delGTP" and "delGDP" respectively, to understand the opening of the nucleotide pockets without the influence of GEFs. As expected, we noted a significant increase in the maximum pocket volumes over $3000\text{\AA}^3$ for both the *apo*-like states, while the minimum pocket volumes stayed close to those observed nucleotide-bound states. This marked increase in the pocket volumes in the nucleotide-free state can be attributed to an increased flexibility in the switch regions and the P-loop (**Figure 2B** and **supplementary figure S6**), which are directly interact with the nucleotides. A more noticeable difference in flexibility (**Figure 2B**) can be seen in the P-loop and the switch I loop (**supplementary figure S6** and **S7**).

The switch region conformation is also found to be reduced by GEF binding (**Figure 2C**), suggesting their role in controlling the nucleotide traffic by modulating the switch-ing regions. We tested this hypothesis by simulating a nucleotide-free and GEF-free form of Rac1 (delGDP and delGTP in **Figure 1B**) and found that the mean pocket volume and fluctuations showed a greater tendency to increase, however, only marginally, as maximum volume was still under $4000\text{\AA}^3$. Next, we measured the pocket volume and the atomic fluctuations in the presence of GEFs in the nucleotide-free state. The volume fluctuations are higher than in the nucleotide-free state without GEFs. However, the volume changes showed significant changes among different GEFs. For example, PREX1 and DOCK2 showed volume changes that are similar in magnitude to *apo*-like states (delGDP and delGTP in **Figure 1B**). TIAM1 and VAV1 showed significantly enlarged nucleotide pockets with maximum volume above $4000\text{\AA}^3$. We hypothesise that such a difference in

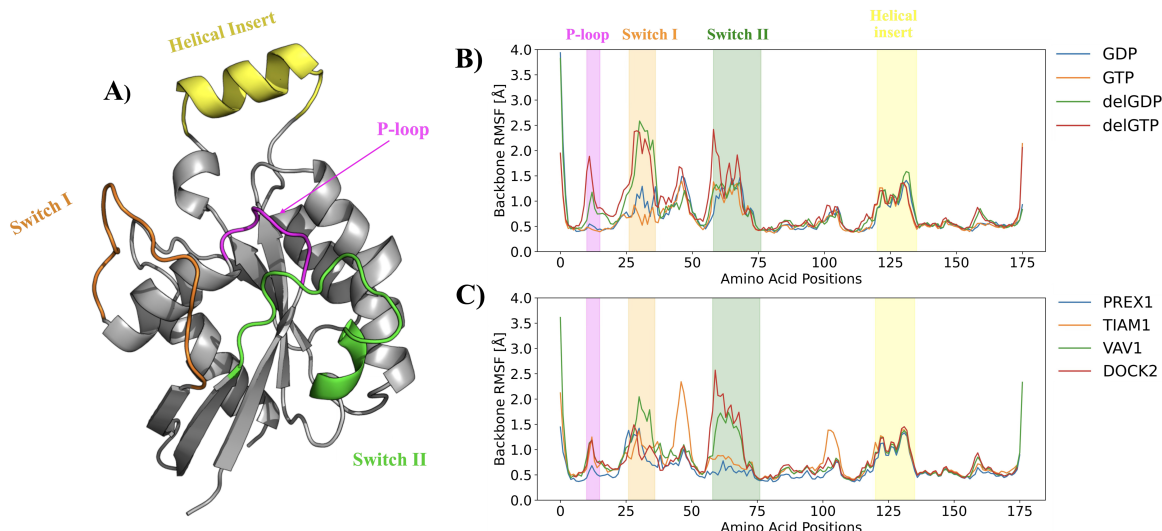


Figure 2: Rac1 domains and dynamics of each region. **A)** A Cartoon representation of human Rac1 (**PDB id:1HE1**). The P-loop region is depicted in the magenta colour. The Switch I region is depicted in Orange Colour, and the Green coloured region is the Switch II region. The helical region is shown in yellow colour. **B)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for the human Rac1 domain to compare the effect of nucleotide binding on the dynamics of various domains. **delGDP** and **delGTP** are systems created by deleting the nucleotides from the Rac1 binding pocket to create corresponding *apo* system. **C)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for the human Rac1 domain to compare the effect of GEF binding on the dynamics of various domains.

pocket volumes by different GEFs suggests that the X-ray structure has captured the Rac1-GEF complexes either just after GDP dissociation (PREX1 and DOCK2) or in a state that was pre-organised for GTP:Mg²⁺ association (TIAM1 and VAV1). As we have already established that pocket volume changes in the nucleotide can be attributed to the P-loop and switch region dynamics, it becomes apparent that GEFs interacting primarily with the switch region can control the nucleotide exchange (**figure 2C**). The nucleotide is seen to enlarge with all GEFs, which is a prerequisite for ejecting GDP, and then perhaps enlarge even more to allow association of GTP and Mg²⁺.

We identified four transient pockets on the Rac1 surface that cannot be located with Rac1 structure alone. The sites P1 and P2 are located in the region lining the switch regions I and II. These sites form the binding site for the GEFs that perturb the switch regions, leading to the opening of these sites. Our pocket volume measurements indicate that the volumes are large enough to accommodate small-molecule binders. In the absence of the GEFs, these sites are too small, and hence, not detected in GEF-free Rac1(**Figure**

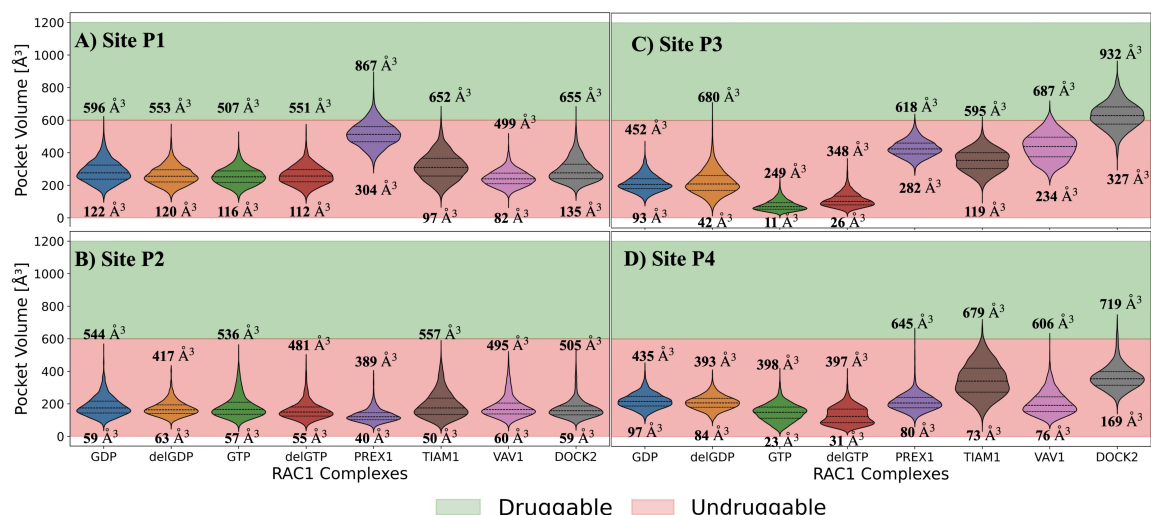


Figure 3: Four binding sites identified using POVME2. **A)** Site P1 is located between switch regions I and II. It appears to be a site similar to KRAS-G12C covalent inhibitors. **B)** Site P2 is located at the bottom of the switch I and II loops at the N-terminal region. **C)** Site P3 is located at the site from where the switch I region. **D)** Site P4 is the only pocket that is not located close to switch I or II regions; however, this site appears to accommodate the ELMO1 protein that assists the DOCK family of GEFs in their function. The threshold for druggability of a pocket was set according to the following references.^{36–38} The numerical values on the violin plots indicate the minimum and maximum pocket volumes measured in the MD trajectories.

1A). Site P1 in Rac1 is similar to the one in K-RAS and H-RAS^{44,45} that is occupied by the covalent inhibitors in the G12C Ras mutant. Sites P1 and P2 are present at a crucial location that could modulate the dynamics of the switch regions. Thus, we hypothesise that one of the consequences of targeting these sites is that it could compete with GEF binding. Alternatively, the molecules occupying these sites could act as molecular glues that stabilise the Rac1-GEF complex, slowing the switching mechanism and thus, the nucleotide exchange. Both scenarios suggest that the modulation perturbs the GEF-mediated downstream signalling (**Figure 3B**). Due to the differences in the binding volumes caused by different GEFs, we anticipate that different Rac1-GEF complexes could be selectively targeted by small molecules of different binding complementarity. We also acknowledge that one of the secondary effects of targeting site P1 and P2 opens up a possibility to occlude GTP binding, albeit non-competitively. According to one of the accepted mechanisms of nucleotide exchange,² GTP binding weakens the GEF binding, which eventually dissociates. By preventing or decreasing the rate of GEF dissociation, GTP binding could be prevented or slowed down. Thereby prolonging the nucleotide-free

state and preventing the downstream signalling. Site P2 is located below Site P1, and also found to marginal increase in the pocket volume in the presence of GEFs (**Figure 3C**). We propose that, owing to the proximity, this could serve as an extension for molecules binding to Site P1 (**Figure 1A** and **supplementary figures S4A and S4B**).

Away from the GEF binding site are located sites P3 and P4 (**Figure 1A**). Site P3 is located towards the N-terminal side of the switch I region. From the X-ray crystallographic information, Residues lining the site P3 are known to interact more specifically with DHR-containing GEFs belonging to the DOCK family.^{21,46} Targeting this site could modulate the switch I region dynamics, which in turn could have an influence on the traffic to the nucleotide-binding site as well as GEF binding. The ELMO1 protein that facilitates the binding of the DOCK family of GEFs is known to interact with residues that form the site P4.⁴⁶ Taken together, we hypothesise that modulation of site P3 and P4 could specifically target and modulate the downstream signalling involving the DOCK family of GEFs.

Accelerating the pocket opening using cosolvent

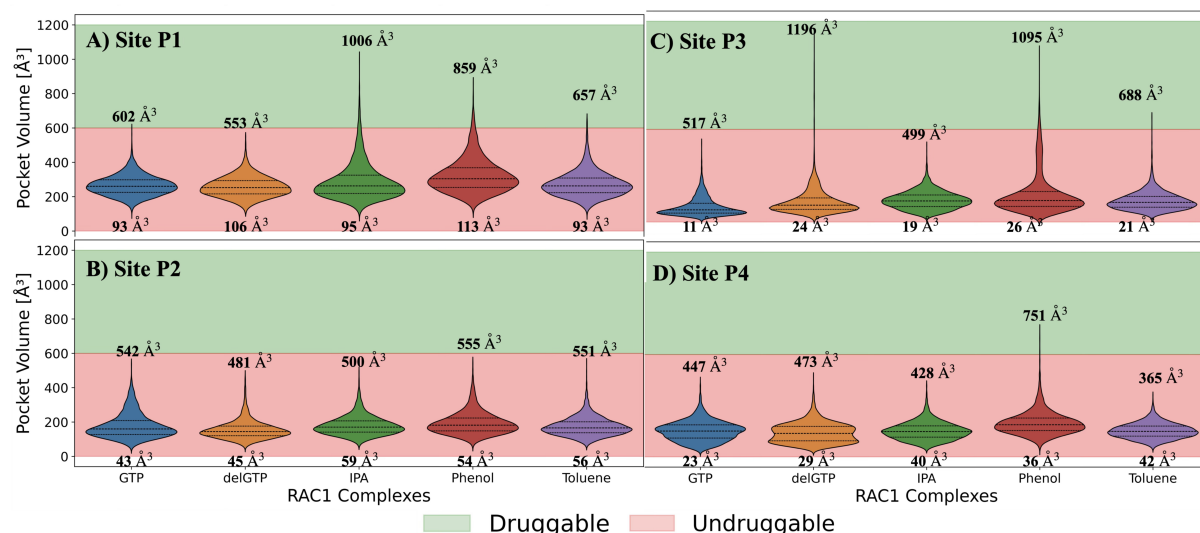


Figure 4: Pocket volume measurements following cosolvent MD simulations for **A)** Site P1, **A)** Site P2, **A)** Site P3, and **A)** Site P4. The numerical values on the violin plots indicate the minimum and maximum pocket volumes measured in the MD trajectories.

Transient pockets open in response to ligand binding, suggesting an induced fit mechanism. However, in the absence of known binders, cosolvents that structurally resemble

drug-like fragments are used as chemical probes to accelerate the opening of binding sites that require ligand binding. In this work, we used three commonly used cosolvents with different hydrophobicities to assess the ability of these transient sites to open up in the absence of GEFs, suggesting their amenability to be targeted by small molecules. The PBD entry with accession number 3TH5 is the only X-ray crystal structure that is crystallised in the active-like conformation bound to GTP analogue, Guanylyl imidodiphosphate (GNP), without any regulatory protein. Rac1 conformation in this case shows that all GEF-induced pockets are in the undruggable state, evident by their small pocket volumes. We performed microsecond timescale MD simulations to check the effectiveness of the chemical probes to open the transient pockets through the induced-fit mechanism. We observe that at three sites, P1, P3, and P4, the pocket volumes cross the threshold value. Therefore, suggesting that these pockets can open large enough to be targeted by small molecules. However, systems (GTP and delGTP) without cosolvent probes remain below the threshold (**Figure 4**) value, confirming their transient nature. This warrants an induced-fit mechanism that is involved in the opening of these pockets. Physiologically, this induced-fit mechanism is observed in response to the receptor pre-organisation step for GEF binding. In the case of site P1, the pocket volume for the reference systems (GTP and delGTP) largely remains below the threshold for druggability. The chemical probes, isopropanol and phenol, can open this site significantly larger than the reference system, attaining a maximum volume of $1006\text{\AA}^3$ and $859\text{\AA}^3$ respectively(**Figure 4A**). While using a completely hydrophobic probe, toluene, the pocket volume increased only marginally (maximum volume measured is $657\text{\AA}^3$). This can be attributed to the phase separation observed for toluene molecules due to its high hydrophobicity, which leads to practically insoluble in water. To prevent such phase separation, it is recommended to use an intermolecular repulsive potential applied to probe molecules.⁴⁷ For site P2, none of the chemical probes were able to increase the pocket volume above the threshold value (**Figure 4B**). Upon careful observation, we note that the site P2 is lined by β -sheets, which form the majority of the site. This leads to a rather flat pocket lined by rigid secondary structural elements, leaving little scope for movements that could increase the

pocket volume. However, we expect this site, P2, to serve as a subpocket for site P1 binders that can be expanded to site P2 to increase the binding affinity.

Site P3 appears to be open spontaneously in *apo*-like (delGTP) with a maximum volume measured is $1196\text{\AA}^3$ (**Figure 4C**). While the pocket volume remained beyond the threshold for GTP-bound Rac1, and with IPA cosolvent. However, phenol and toluene could push the maximum volume beyond the threshold, wherein with phenol the maximum volume measured was $1095\text{\AA}^3$, which is similar magnitude to delGTP. The residues that make up site P3 appear to be mainly aromatic and aliphatic, which makes this pocket a hydrophobic patch (**Figure 1A**). Therefore, the hydrophilic cosolvent, IPA, was unable to sample the open state of the pocket, whereas the hydrophobic probes like phenol and toluene did sample the open state of the pocket, wherein the pocket volume measures crossed the druggability threshold. Site P4 appears to be the smallest among the four transient pockets identified in this work. We observed that only in the phenol cosolvent system the pocket volume crossed the druggability threshold with a maximum volume of $751\text{\AA}^3$ (**Figure 4D**). Upon closer inspection of the binding site residues, we observe a good balance of hydrophobic and charged residues, owing to which phenol, which comprises a hydrophobic ring with a hydrophilic hydroxylic group, was able to open to the pocket beyond the threshold value. In summary, we note that three out of four pockets are indeed transient in nature, which can be open in an induced-fit manner. This is exemplified by the GEF-binding and the cosolvent probe molecules. We did choose the GTP-bound conformation of the Rac1 protein in which all four pockets reported the pocket volumes below the threshold value (**Figure 3**). This way we could assure that in the starting conformation all pockets could be labelled as practically undruggable, giving an unbiased starting point to test the ability of the cosolvent probes to open the pockets.

Switch region....

To understand the protein dynamics that are involved in the opening of these transient pockets, we extracted the essential dynamics using the principal component analysis of Rac1 following microsecond-scale cosolvent MD simulations. The purpose of such

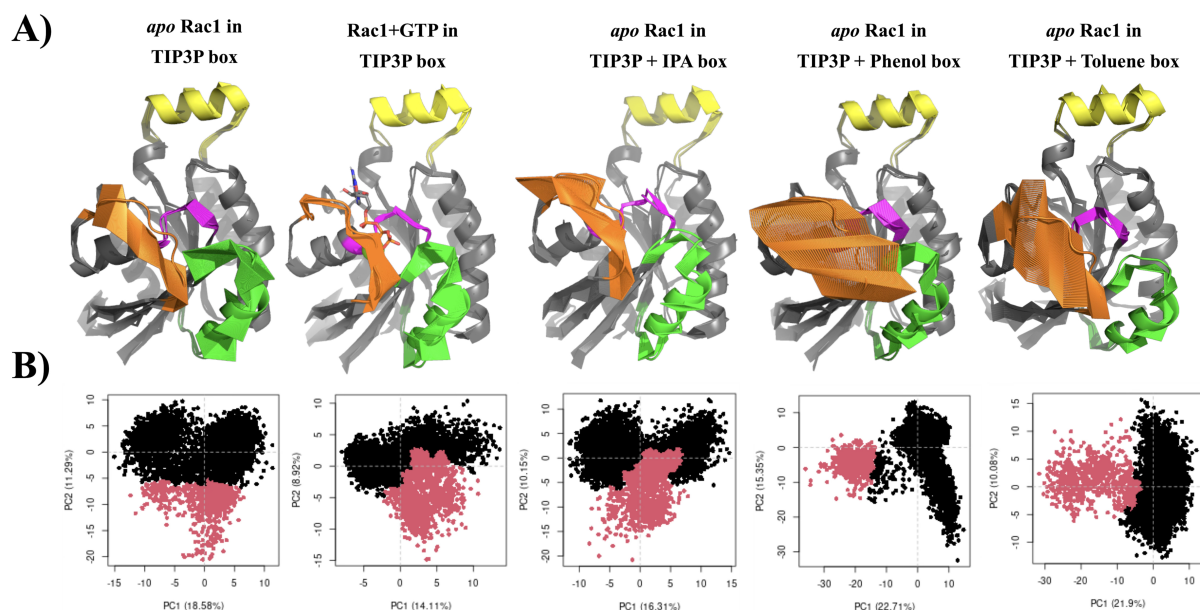


Figure 5: **A)** Essential Dynamics of Rac1 protein to explain the opening and closing of the transient pockets induced by the cosolvent probes. Magenta colour indicates the P-loop region. Orange colour indicates switch I region. Green colour indicates switch II region. Yellow colour indicates helical insert. IPA is isopropanol. **B)** Identification of Rac1 conformational clusters in various water and cosolvent systems based on essential dynamics using the first two PCs.

long-time-scale simulations with and without cosolvents is to evaluate the possibility of spontaneous opening of the pockets, and to confirm the induced-fit mechanism needed for opening the pockets. The systems that were simulated without cosolvents are treated as the reference systems for comparing the cosolvent-induced pocket opening. In RAC1, like all other small G proteins, the switch regions show higher conformational flexibility. This flexibility is central to the switching mechanism for small G proteins that allows GEF and GAP binding, which is a prerequisite for nucleotide exchange and GTP hydrolysis, respectively. The essential dynamics extracted from the dominant modes of motions by employing principal component analysis also confirm the large dynamical flexibility of the switch regions. Therefore, we embarked on understanding how the switch region dynamics could play a role in the opening and closing of these transient pockets. GTP and delGTP systems that were simulated without cosolvent show that switch regions show maximum movements in the Rac1 protein (**Figure 5A**). A significant ordering of switch I, and the P-loop is observed in the presence of GTP and Mg^{2+} ion. This is attributed to the strong electrostatic interactions shown by the phosphate group through the Mg^{2+} ion,

which interacts with the switch I residues. The Rac1 system simulated on the cosolvent box with nucleotides shows the largest dynamics in the switch I region (**Figure 5A**), while the maximum pocket volumes are also observed in these conditions(**Figure 4**). We do not observe any other structural domain in Rac1 with dynamical flexibility as large as the switch I region. As we compare the effectiveness of the cosolvent probes in opening the transient pockets, phenol and toluene show maximal efficiency, in which the PCA-based clustering (**Figure 5B**) using the first two principal components shows distinguishable conformations between open and closed states. Since phenol and toluene are fragments that frequently appear in drug-like molecules, they are more efficient in exhibiting induced-fit opening of the pockets, a behaviour that can be expected from drug-like molecules.

So far, we have observed that most of the transient pockets in Rac1 open to accommodate the GEFs required for the nucleotide exchange. Hence, we found it prudent to analyse the energetic contribution of the Rac1 residues that are involved in GEF binding. Furthermore, we attempt to locate how many amino acid residues important for GEF binding also make up the four transient sites, and thus, the per-residue contribution can indicate the binding hot spots for small molecules. As described earlier, there is a dominant role of the switch regions in GEF binding that leads to dissociation of GDP and association of GTP. This nucleotide exchange, catalysed by the GEF, necessitates their interaction with the switch regions, thus modulating their conformation such that it facilitates this exchange. The sites P1, P2 and P3 are located in the vicinity of the GEF binding region on Rac1. A thorough analysis of the X-ray crystallographic structures suggests that important amino acid residues in Rac1 involved in GEF binding and activation are located in the switch I and II regions.¹⁸⁻²¹

Site P1 is located in the cavity that forms between switch I and II, which is the region where all the GEFs bind to perform their functions(**Figure 7**). Seven residues (T35, V36, F37, D38, W56, and D57) lining this site contribute significantly to GEF as shown by MMGBSA per residue decomposition. From the crystallographic data, the backbone amides of T35 and V36, which are located on the switch I region, show

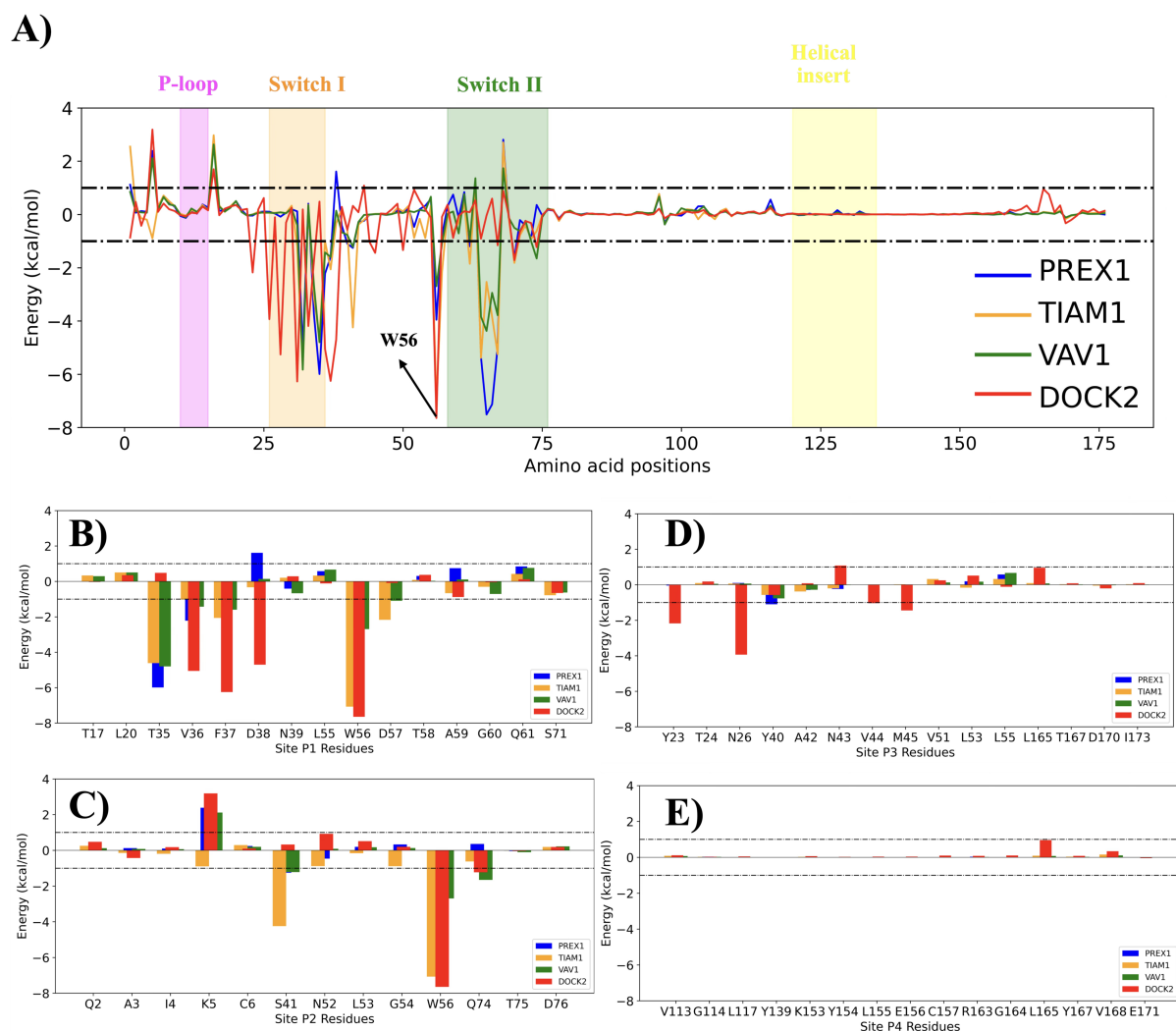


Figure 6: A) Per residue contribution to the effective binding affinity of the Rac1 residues to the GEFs as computed by MMGBSA. The P-loop region is depicted in the magenta colour. The Switch I region is depicted in Orange Colour, and the Green coloured region is the Switch II region. The helical region is shown in yellow colour. The residue W56 is marked due to its biological importance in the GEF binding B) Per residue contribution from Site P1 residues. C) Per residue contribution from Site P2 residues. D) Per residue contribution from Site P3 residues. E) Per residue contribution from Site P4 residues. The "dot-dash" line indicates an arbitrary cutoff placed at ± 1 kcal/mol, which suggests that energy values outside this region contribute significantly to the binding of GEFs.

hydrogen bonding interactions with conserved GEF residues, such as E56(PREX1)¹⁸ and E201(VAV1).²⁰ F37 and D38, which are also located on the switch I region, show pronounced favourable interactions with DOCK2 residues (**Figure 6B**). W56 is a key determinant of selectivity across both DOCK and DH-PH GEF families. In Rac1, W56 is a bulky residue that inserts into a pocket formed in the GEFs, and its presence generally supports activation by Rac-specific DOCKs. W56 is also the only residue that forms part

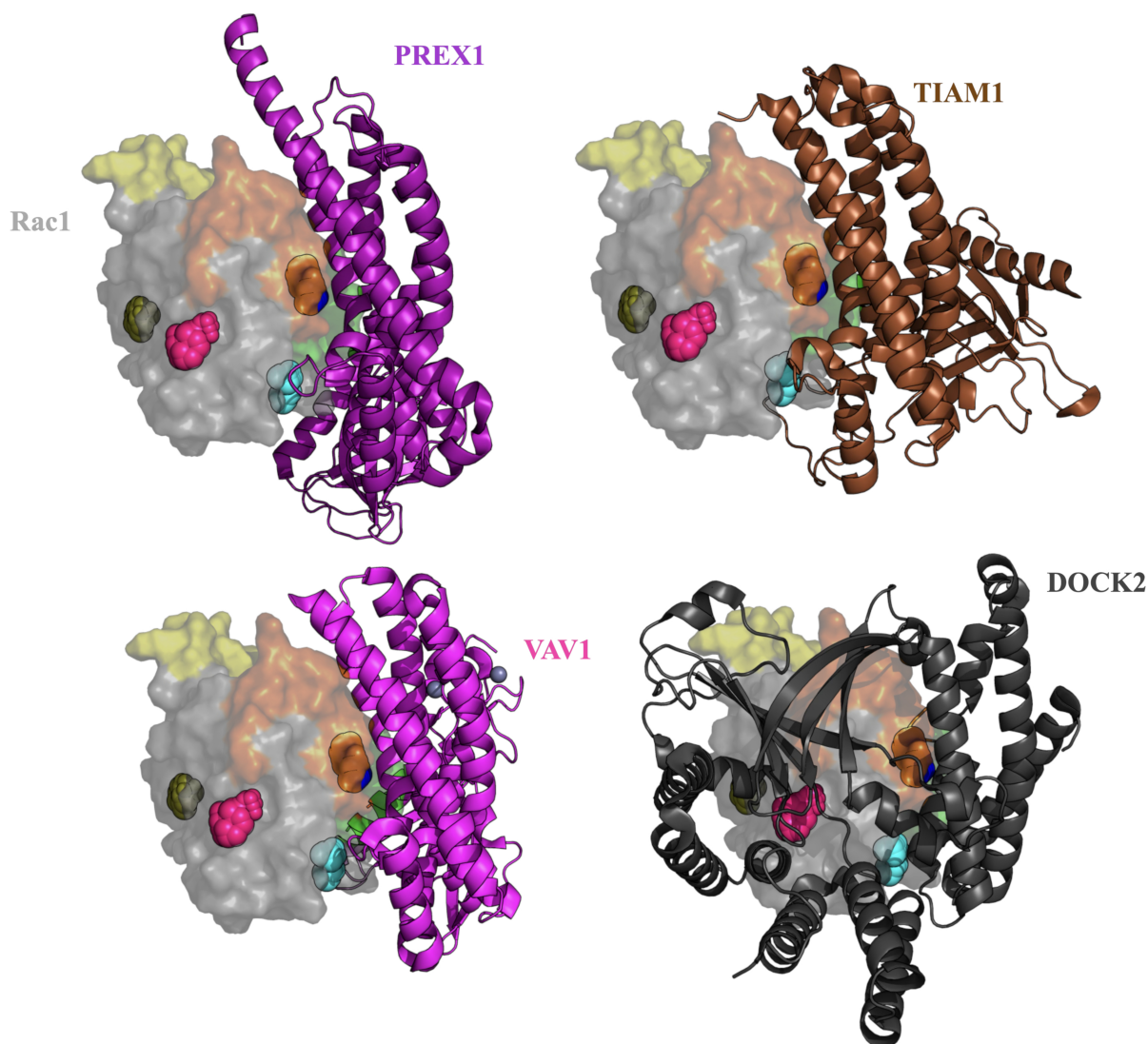


Figure 7: Rac1-GEF complexes with four transient sites illustrated to understand the orientation of various GEFs on the Rac1 surface. Rac1 is shown in a grey surface representation. All GEFs are shown as a cartoon representation. The transient pockets are represented as the Speres: Blue spheres show site P1, Cyan spheres show site P2, magenta spheres show site P3, and Asparagus spheres show site P4.

of Site P1 and P2; therefore, designing small molecules that increase interaction with it could lead to competition with GEF binding. In summary, the residues that form Site P1 and P2 show a significant number of interactions with GEF proteins. Furthermore, we also observe figure 6 that there are residues that selectively bind to different GEFs that could be targeted for GEF-specific modulations. For example, F37 and D38 show markedly greater interaction with DOCK2 than other GEFs used in this study; therefore, small. Molecules that show stronger interactions with these residues could selectively compete with the DOCK2 protein. In case of sites P3 and P4, residues interact with

GEFs(**Figure 6D** and **6E**). However, a few residues in site P3 show modest interaction with DOCK2, which makes this pocket more amenable to targeting and modulating Rac1-DOCK2 interaction(**Figure 7**).

Conclusion

The computer simulations reported in this work reveal that Rac1 harbours four transient, GEF-sensitive pockets whose opening is tightly coupled to switch-region dynamics. We tested the opening of these transient pockets using cosolvent probes that suggest an induced-fit mechanism. Sites P1 and P2 are located in the regions that form the GEF docking interface. While sites P3 and P4 map to regions engaged by the DOCK-family of GEFs and the ELMO1 cofactor. Pocket-volume fluctuations across nucleotide- and GEF-bound states, measured by microsecond-scale cosolvent MD, demonstrate that several of these pockets cross the druggability threshold, and collapse to undruggable sizes in the absence of regulatory partners. This behaviour confirms their transient nature and warrants an induced-fit mechanism that triggers the opening of the pockets. Taken together, these findings suggest that Rac1, which was considered undruggable, can in fact be modulated using small molecules that target the transient pocket we identified in this work. We strongly believe that this work offers new opportunities to target specific Rac1-GEF complexes with small molecules to modulate GEF-specific pathways.

Acknowledgement

Supporting Information Available

The files associated with this project can be downloaded from **GitLab orbitMD**. List of supplementary figures

- **Figure S1:** Pearson's Correlation for the volume fluctuations among different pockets and the effect of nucleotide binding.
- **Figure S2:** Pearson's Correlation for the volume fluctuations among different pockets and the effect of GEF binding.
- **Figure S3:** Scree plot to compute the cumulative variance explained by principal components.
- **Figure S4:** Analysis of pocket volume changes by combining pockets P1 with P2, and P3 with P4.
- **Figure S5:** Time-evolution of RMSD of backbone atoms from Full Rac1 protein.
- **Figure S6:** Time-evolution of RMSD of backbone atoms from the P-loop region.
- **Figure S7:** Time-evolution of RMSD of backbone atoms from the switch I region.
- **Figure S8:** Time-evolution of RMSD of backbone atoms from the switch II region.

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